

PIAN ROSA, Italy

WMO: 160520

Lat: 45.935N

Lon: 7.706E

Elev: 11444

StdP: 9.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-7.5	-25.7	2.0	10.5	-22.4	2.5	14.1	58.6	5.9	55.3	6.9	25.4	60	1.070

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	48.8	40.0	46.6	39.0	44.7	37.7	42.7	46.7	41.0	44.8	39.4	43.1	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
40.7	57.7	44.2	38.8	53.7	42.6	37.0	49.9	41.1	19.9	46.7	18.9	44.6	17.9	43.1	48.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 47.3	40.8	34.2	DB	-14.3	52.1	4.5	4.0	-17.5	55.0	-20.1	57.3	-22.6	59.6	-25.8	62.5	(4)
(5)			WB	-13.7	45.1	4.0	1.8	-16.6	46.4	-19.0	47.4	-21.2	48.4	-24.1	49.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	11.0	11.6	13.4	18.9	25.4	31.9	36.8	36.4	31.0	26.7	17.7	13.7
	DBStd	11.72	8.40	8.41	7.98	6.34	6.12	7.09	5.60	5.44	6.27	6.43	8.72	8.35
	HDD50	9879	1210	1076	1134	934	762	544	410	423	569	723	968	1126
	HDD65	15354	1675	1496	1599	1384	1227	994	875	888	1019	1188	1418	1591
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.4	18.4	15.9	15.2	13.1	12.2	10.7	10.1	9.2	10.4	12.6	15.6	17.5
Precipitation	PrecAvg	35.9	2.2	2.2	2.3	3.3	4.4	3.4	2.8	2.9	3.1	3.5	3.4	2.4
	PrecMax	49.6	4.8	5.2	4.9	6.4	8.5	6.8	6.6	5.6	8.1	8.4	8.5	6.8
	PrecMin	25.5	0.4	0.4	0.3	1.1	2.2	1.6	1.0	1.5	0.7	1.3	0.8	0.2
	PrecStd	5.9	1.3	1.3	1.2	1.6	1.8	1.3	1.1	1.0	1.9	1.9	2.2	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	33.5	32.4	35.5	44.5	50.7	51.9	50.9	46.6	39.5	35.8	30.5
		MCWB	20.2	23.6	23.2	27.8	35.3	38.1	42.1	41.6	38.1	30.9	24.1	20.5
	2%	DB	26.7	29.0	28.3	32.1	39.9	46.8	49.2	48.0	43.5	37.7	32.3	27.8
		MCWB	19.4	20.9	21.3	25.8	32.2	37.8	40.9	40.4	36.0	29.3	23.0	19.7
	5%	DB	24.1	25.2	25.3	29.7	36.3	44.4	46.8	45.8	41.3	36.0	30.2	25.7
		MCWB	18.1	18.1	19.3	24.1	29.9	37.3	39.5	38.9	33.8	28.2	22.6	19.0
	10%	DB	21.6	22.6	23.2	27.1	33.8	41.4	44.8	43.8	39.3	34.3	28.1	23.5
		MCWB	16.6	16.5	17.8	22.7	28.2	35.4	37.9	37.5	32.7	26.8	21.7	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	24.8	26.7	30.1	37.0	43.7	44.9	44.8	41.4	35.4	28.6	25.3
		MCDB	25.9	30.3	30.0	32.7	42.8	47.1	48.7	48.2	44.5	37.3	31.2	25.9
	2%	WB	21.7	21.7	22.8	27.5	33.8	40.5	42.9	42.5	38.7	33.0	26.5	22.8
		MCDB	24.6	26.5	26.1	30.2	38.0	44.4	47.3	46.2	41.8	35.5	28.2	24.5
	5%	WB	19.5	19.4	20.8	25.3	31.3	38.6	41.2	40.7	36.7	31.2	24.8	20.8
		MCDB	22.7	23.2	24.3	28.1	35.2	42.4	45.4	44.1	39.7	33.8	27.6	23.2
	10%	WB	17.4	17.4	19.0	23.3	29.4	36.5	39.4	38.9	34.8	29.4	23.1	18.8
		MCDB	21.1	21.5	22.1	25.9	32.5	40.4	43.4	42.5	37.8	32.2	26.4	22.1
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.6	8.8	8.9	9.1	9.4	9.7	8.8	7.8	7.3	7.6	8.1
		MCDBR	8.6	9.5	9.4	10.7	11.4	11.8	11.8	10.1	8.8	7.7	7.8	7.7
		MCWBR	7.0	7.8	7.9	9.6	10.4	11.4	11.3	11.1	11.5	8.6	6.6	6.6
	5% WB	MCDBR	8.4	9.9	8.9	10.4	10.7	11.3	11.2	9.7	8.4	7.1	6.9	8.1
		MCWBR	8.5	9.6	8.6	10.1	10.3	11.5	11.4	11.3	11.3	9.1	7.6	8.7
Clear-Sky Solar Irradiance	taub	0.182	0.196	0.223	0.251	0.263	0.270	0.268	0.261	0.248	0.231	0.204	0.180	
	taud	2.429	2.397	2.376	2.422	2.492	2.502	2.505	2.538	2.571	2.597	2.545	2.449	
	Ebn at Noon	314	325	326	322	319	316	315	315	313	308	302	304	
	Edh at Noon	21	26	32	34	33	33	32	30	26	22	18	18	
All-Sky Solar Radiation	RadAvg	485	760	1120	1401	1607	1906	1922	1668	1293	831	489	402	
	RadStd	57	100	149	182	138	154	148	124	104	97	73	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (102 neighbors)	N/A	N/A	N/A	+1.26	+0.72	+0.54	N/A	N/A	+101	+29

Nomenclature: See separate page